ENSIA - Machine Learning Project

Spring 2025 - 2026

Forecasting and Identifying Global Export Opportunities for Algerian Exporters

1. Project Overview

Algeria's export economy remains highly concentrated in hydrocarbons; in 2023, oil and gas accounted for approximately 92% of total exports, with overall export revenues estimated at around $50–55 billion, while non-hydrocarbon exports represented only a small share of the export basket despite gradual policy efforts to expand them. This strong dependence exposes the national economy to volatility in global energy prices and reinforces the strategic urgency of diversification. In recent years, national economic policy (under the directives of President Abdelmadjid Tebboune) has placed significant emphasis on increasing non-hydrocarbon exports, improving export competitiveness, and supporting productive sectors to strengthen the national economy.

Within this context, the proposed project aims to develop an end-to-end machine learning system that identifies, analyzes, and forecasts international export opportunities for Algerian exporters across all economic sectors (agriculture, industry, and services). The system will leverage and integrate multiple international trade data sources, including UN Comtrade, WTO statistics, ITC Trade Map, and World Bank trade databases, which provide detailed export and import values by product, country, and time. By analyzing global demand, identifying untapped markets, discovering high-demand products per sector and country, and forecasting trade trends, the project will generate data-driven insights to support the Algerian Chamber of Commerce and Industry (CACI), the Ministry of External Commerce, exporters, and policymakers in achieving export diversification, increasing non-hydrocarbon exports, and enhancing Algeria's long-term economic resilience and global trade positioning.

The system will combine multiple machine learning problem types, including: Clustering (market and product segmentation), Classification (export opportunity detection), Forecasting (trade trend prediction). Optionally, an interactive dashboard will be developed to visualize insights and support real-world decision-making for export promotion.

2. Objectives

1. Collect and integrate international trade datasets from multiple free sources (UN Comtrade, WTO, Trade Map, World Bank, etc.)
2. Analyze global trade flows by product, country, and sector
3. Identify promising international markets for Algerian export products
4. Detect globally high-demand products that Algeria could potentially export
5. Engineer meaningful economic and trade features (growth rates, market share, demand indicators, diversification metrics, etc.)
6. Apply clustering techniques to group countries, products, and sectors based on trade patterns
7. Develop classification models to detect high-opportunity export markets
8. Build forecasting models to predict trade volume and value trends by product and country
9. Design a visualization dashboard to support exporters and the Algerian Chamber of Commerce and Industry in strategic planning
10. Ensure ethical data usage and respect privacy and data sensitivity constraints

3. Methodology

Step 01. Data Collection

- International trade data (exports/imports by product and country)
- Economic indicators (GDP, trade growth, market size)
- Algerian export data (if available from CACI, Ministry of External Commerce or public sources)
- Sectoral and product-level trade statistics

Possible data sources:

- UN Comtrade (global trade database) https://comtrade.un.org/
- WTO trade statistics https://data.wto.org/
- ITC Trade Map (aggregated trade flows) https://trademap.org/
- World Bank open economic data https://data.worldbank.org/ https://wits.worldbank.org/

All datasets will be publicly available and compliant with data privacy and security considerations.

Step 2. Data Preparation & Integration

- Cleaning and harmonizing multi-source datasets
- Handling missing values and inconsistencies
- Data normalization and scaling
- Feature engineering:
  - Export growth rate
  - Global demand index
  - Market penetration ratio
  - Trade balance indicators
  - Product diversification metrics

Exploratory Data Analysis (EDA) will include:

- Trade flow visualizations
- Sector demand analysis
- Country-product heatmaps

Step 3. Model Development

The project will address multiple machine learning tasks:

a) Clustering Tasks

- Group countries based on import demand patterns
- Cluster products based on global demand trends
- Identify similar export markets

b) Classification Tasks

- Classify country-product pairs into:
  - High export opportunity
  - Medium opportunity
  - Low opportunity

c) Forecasting Tasks

- Predict future trade volume and value trends
- Forecast demand for specific products and sectors
- Analyze long-term export growth potential

Step 4. Evaluation

Different evaluation strategies will be used depending on the task:

- Forecasting: Error metrics (MAE, RMSE, MAPE)
- Classification: Accuracy, Precision, Recall, F1-score
- Clustering: Silhouette Score, Davies-Bouldin Index
- Economic relevance validation (interpretability of results)

Step 5. Validation

- Testing on unseen time periods and countries
- Sensitivity analysis across sectors and products
- Comparison of insights across multiple data sources
- Real-world validation using known Algerian export trends

4. Expected Deliverables

- Integrated multi-source (at least two) international trade dataset (UN Comtrade, WTO, Trade Map, etc.)
- Documented Jupyter notebooks (data preprocessing, feature engineering, modeling)
- Machine learning models for clustering, classification, and forecasting tasks
- Export opportunity ranking system (by country, product, and sector)
- Interactive visualization dashboard using Grafana or similar professional tools (e.g., Power BI, Superset, Kibana, Metabase), only one dashboard to keep it simple and easy
- Deployed data pipeline connected to the dashboard (API, database, or CSV ingestion)
- Final technical report explaining methodology, models, and economic insights
- Final presentation and live system demonstration

5. Success Criteria

The system should:

- Correctly identify promising international markets for Algerian exporters
- Discover high-demand global products aligned with Algerian export potential
- Provide accurate and interpretable trade forecasts
- Produce meaningful clusters of countries and sectors based on trade behavior
- Deliver actionable insights useful for CACI and export-support institutions
- Maintain strong model performance across multiple sectors and datasets

- Demonstrate practical usefulness for improving export strategy and economic planning

6. Demonstration

During the demonstration session, students will present the complete end-to-end machine learning system, including:

- Multi-source international data integration
- Exploratory data analysis and economic interpretation
- Clustering of countries, products, and sectors
- Classification of export opportunities
- Forecasting of trade volume and demand trends
- Deployment of a professional visualization dashboard

Students must demonstrate an interactive dashboard built using:

- Grafana (recommended), or
- Other professional visualization tools such as Apache Superset, Metabase, or Kibana

The dashboard should allow stakeholders (CACI experts, exporters, policymakers) to:

- Explore export opportunities by country, sector, and product
- Visualize global demand trends and trade flows
- Monitor predicted export growth and market potential
- Identify priority international markets for Algerian exporters
- Compare historical vs forecasted trade indicators

7. Pedagogical Value

Including Grafana or similar tools will help students:

- Learn real-world ML deployment workflows
- Work with large economic datasets (not toy datasets)
- Develop data engineering and dashboarding skills
- Understand how ML supports economic policy and export strategy
- Gain industry-level competencies aligned with government and institutional projects.